Sainath Punchbhai

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SUMMARY

Robotics-focused engineer with hands-on experience building robotic arms, hexapods, drones, and autonomous systems. Experienced across the robotics stack—from CAD, electronics, embedded control to ROS2, URDF/Xacro, MoveIt2, RViz, Gazebo, and manipulation. National-level competition winner and runner-up, seeking opportunities in robotics systems, manipulation, and embedded robotics.

EDUCATION

B.M.S. College of Engineering, Bengaluru **2024 - Present**
Electronics and Communication Engineering
- **Relevant Coursework**: Digital Electronics & Logic Design, Analog Circuits, Microprocessors & Embedded Systems (ARM), Instrumentation & Sensors, HDL(Verilog)-Programming.

SKILLS

Programming: Python, C/C++
Developer Tools: ROS 2, MoveIt 2, Gazebo, RViz, URDF/Xacro, Linux, SOLIDWORKS, Fusion 360, KiCad, EasyEDA, ArduPilot-Mission Planner, Arduino.
Domains Explored: Robotics, Manipulation, Motion Planning, Embedded Systems, Autonomous Systems, UAVs, Computer Vision, Electronics & PCB Design, CAD & Prototyping, Sensor & Actuator Integration, Microcontrollers, Digital Circuits.

PROJECTS

Dummy - ROS 2 Robotic Arm & Manipulation System
- Converted the custom **CAD assembly** into URDF/Xacro, creating a ROS2 robot description and simulation-ready model.
- Built a ROS2 control and visualization stack using **RViz** and integrated **MoveIt 2** for **inverse kinematics, motion planning and trajectory execution**.
- Developed a simulated **manipulation environment in Gazebo**, integrating the robot model with ROS2 for **physics-based testing and digital-twin development**.
- Implemented and tested **pick-and-place workflows** for small objects, iterating on robot geometry, joint configuration and **motion planning**.

Pete - Six-Legged Hexapod Robot
- Designed and developed a six-legged (hexapod) robot, creating the complete mechanical design using **CAD software.**
- Selected and **integrated hardware** components, managing multiple servo motors and efficient power distribution.
- Engineered leg mechanisms, joint placements, and assembly structure to achieve **smooth and coordinated locomotion**.
- Gained hands-on experience in **3D printing**, prototyping, and iterative design optimization for reliable mechanical performance.

PID Line Follower Robot
- Designed and developed a PID-controlled autonomous line follower with a **custom CAD-designed chassis** for improved stability.
- Utilized a **multi-IR sensor array** for precise line detection, high-resolution error correction, and reliable path tracking.
- Implemented and **tuned PID control** to minimize oscillations and achieve smooth navigation.

EXPERIENCE

Robotics Club BMSCE **Dec 2024 - Present**
Electronics Design & Processing(EDP) Member
- Developed a **6-DOF robotic arm** for the **International Rover Challenge**, focusing on precise manipulation and control.
- Designed and integrated **electronic systems** for robotics projects, including **robotic arms, balance bots, and line-followers.**

AeroBMSCE **Jul 2025 - Present**
Avionics Team Member
- Designed, fabricated, and tuned autonomous UAV for **SAE INDIA Aerothon 2025 and SAE India ADDC 2026**.
- **Integrated** flight controllers, sensors, communication modules, and embedded systems for **autonomous flight and payload operations.**

Toastmasters International - Republic of Toastmasters **Aug 2025 - Present**
Vice President Membership
- **Led membership, branding, and public-relations initiatives** while coordinating club activities and supporting organizational growth.
- Completed Level 4 – Dynamic Leadership Path, strengthening **leadership, communication, and project-management skills**.

BMSCE IEEE PES & Sensors Council **Nov 2024 - Nov 2025**
SAC - Students Activities Committee
- **Coordinated EcoForge**, a 24-hour H**ardware Hackathon** at PhaseShift-25, managing logistics, execution, and participant coordination.
- **Organized technical workshops and events** focused on electronics, embedded systems, and sensor technologies.

ACHIEVEMENTS

- **RUNNERS UP** of National Level **SAE INDIA AEROTHON 2025.**
- **Winners** of **SAE INDIA Autonomous Drone Development Competition 2026** under the **Unique Design Category.**
- **Qualified to Stage-2** (Hardware) of **E-Yantra Robotics Competition 2025**.
- Secured **TOP-5** in the 24-hour **Hardware Hackathon** hosted during PhaseShift 2024.